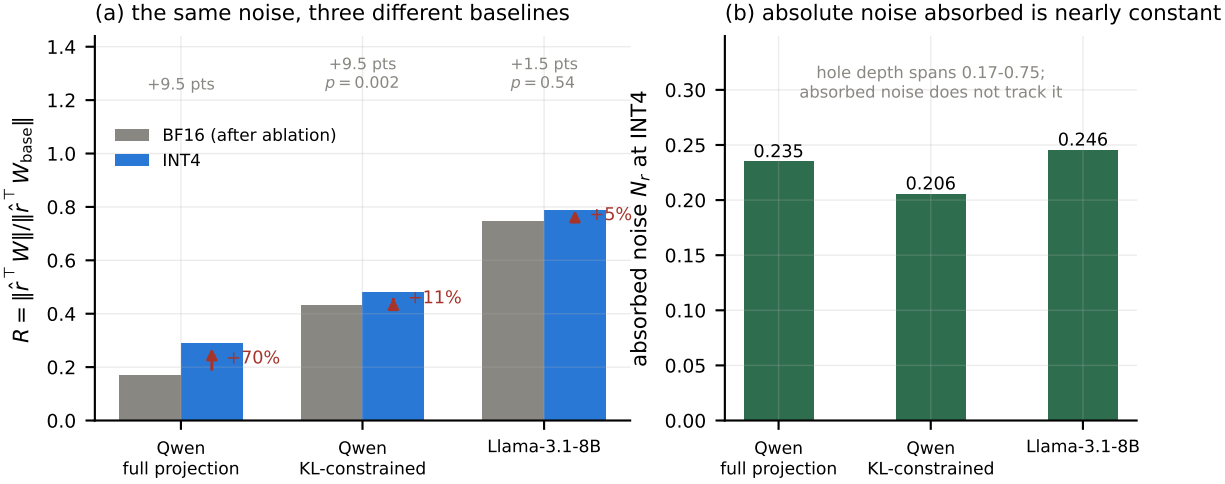
(a) the same noise, three different baselines
+9.5 pts
+9.5 pts
$p = 0.002$
+1.5 pts
$p = 0.54$
BF16 (after ablation)
INT4
$R = \|\hat{r}^\top W\| / \|\hat{r}^\top W_{\text{base}}\|$
+70%
+11%
+5%
Qwen
full projection
Qwen
KL-constrained
Llama-3.1-8B
(b) absolute noise absorbed is nearly constant
hole depth spans 0.17-0.75;
absorbed noise does not track it
absorbed noise N_r at INT4
0.235
0.206
0.246
Qwen
full projection
Qwen
KL-constrained
Llama-3.1-8B